

The eleven capitals at stake when AI operates in high-risk sectors.

What is at risk — for those who operate the system
and for those who receive its decisions.

When AI fails in high-risk sectors, it does not destroy just one thing.

It destroys simultaneously across multiple forms of value — for both operators and recipients.

01 HUMAN Professional judgment and beneficiary dignity	02 SOCIAL Community trust and institutional fabric
03 CULTURAL Institutional legitimacy and recognized authority	04 ECONOMIC Resources, quantifiable risk and market access
05 POLITICAL Capacity to supervise, regulate and influence	06 ETHICAL Moral trust — of the professional and of society
07 TECHNOLOGICAL Technical assets and standards as infrastructure	08 INSTITUTIONAL Verifiable norms and public policy
09 NATURAL ★ Ecosystems, vital cycles and biodiversity	10 INTELLECTUAL ★ Professional knowledge, innovation and collective judgment
11 MANUFACTURED ★ Physical infrastructure: networks, roads, plants, hospitals	

★ **Natural, Intellectual and Manufactured: the three most overlooked capitals — and the hardest to recover.**

Each capital is analyzed from two perspectives: who operates the system and who receives its decisions.

HUMAN

WHAT IT IS

Human capital encompasses the accumulated knowledge, technical skills, professional judgment and decision-making capacity of those who operate in a sector. It is the most direct asset a professional puts at stake every time a decision is made.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

The professional who uses AI without verifiable criteria loses the ability to distinguish when to trust the system and when not to. Their expertise becomes a vulnerability: they assume responsibility for decisions they cannot audit.

WHO RECEIVES THE DECISION

The patient or citizen assumes the professional exercised their judgment. If the system failed and that judgment could not be exercised because no one could interrogate it, the beneficiary suffers the harm without knowing its origin and without being able to contest it.

AFFECTED SECTORS

T4 — CRITICAL

| Hospital triage · Traffic control · Emergency management

T3 — HIGH RISK

| Clinical diagnosis · Clinical laboratory

Personnel selection · Educational assessment

WHY IT MATTERS — BOTH PERSPECTIVES

A physician who cannot assess the limits of a diagnostic support system does not detect when the algorithm is failing. The patient receives an incorrect diagnosis without knowing it. The physician signs off without knowing it. When the failure is discovered, the physician faces legal consequences and the patient has already borne the harm. The human capital of both was damaged by the same absence of standards — but in completely different ways.

SOCIAL

WHAT IT IS

Social capital encompasses trust networks, cooperation between actors, institutional cohesion and community fabric. It includes citizens' trust in the institutions that govern them.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

The professional or institution operating AI without standards cannot demonstrate diligence when something fails. The trust of their professional networks is exposed by a failure they cannot explain.

WHO RECEIVES THE DECISION

When the beneficiary discovers they were affected by an unsupervised system, they lose not just confidence in the specific institution — they lose confidence in the entire system. A community that learns the algorithm operates against it disconnects from the State.

AFFECTED SECTORS

T4 — CRITICAL

| Emergency management · Water resource management

T3 — HIGH RISK

| Judicial system · Social protection

Credit and banking · Public safety · Migration

WHY IT MATTERS — BOTH PERSPECTIVES

A family that receives a credit denial from a biased algorithm does not file a technical complaint — they stop trusting the bank, the regulator and the financial system. The official who operated the system cannot explain the decision either. The social capital of both was eroded by the same failure: no one had a tool to interrogate the system.

CULTURAL & SYMBOLIC

WHAT IT IS

Cultural capital encompasses legitimacy and socially recognized authority. Symbolic capital is the accumulated prestige that makes others trust. For beneficiaries, it is the expectation of being seen as people, not as data.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

A professional association or regulator without a technical position on AI loses legitimacy when a system fails publicly. They cannot defend their members or reclaim their supervisory role. Their cultural capital is destroyed at the worst possible moment.

WHO RECEIVES THE DECISION

The beneficiary harmed by an unsupervised system loses more than the specific outcome: they lose the certainty that institutions are competent to protect them. That symbolic damage — feeling invisible to the system — is harder to repair than material harm.

AFFECTED SECTORS

T4 — CRITICAL | Hospital triage · Emergency management

T3 — HIGH RISK | Judicial system · Clinical diagnosis

Educational assessment · Public procurement

WHY IT MATTERS — BOTH PERSPECTIVES

A student failed by an algorithm no one can explain does not just lose a grade: they lose trust that the educational system sees them as a person. The institution, in turn, loses the authority for its decisions to be respected. The symbolic capital of both was damaged by the same opacity.

ECONOMIC

WHAT IT IS

Economic capital encompasses financial resources, market value, access to financing and the ability to quantify risk. It includes the real cost of failures and access to regulated markets.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

An organization deploying AI without standards assumes unquantified economic risk: losses from failures, insurance premiums without metrics, exclusion from regulated markets and legal liability without traceability to mount a defense.

WHO RECEIVES THE DECISION

The beneficiary who receives an incorrect decision about their credit, pension or service access loses real resources in ways that are frequently irreversible in the short term: missed opportunities, accumulated debt, and denied access cannot be fully restored by a later correction.

AFFECTED SECTORS

T4 — CRITICAL

| Energy infrastructure · Telecommunications

T3 — HIGH RISK

| Credit and banking · Pensions

Public procurement · Agriculture · Personnel selection

WHY IT MATTERS — BOTH PERSPECTIVES

An elderly person whose pension was miscalculated by a robo-advisor has no time to recover that loss. The entity that operated the system cannot prove diligence if there is no traceability. The economic harm to the beneficiary is immediate; the institution's harm is deferred but equally real: legal liability and regulatory exclusion.

POLITICAL

WHAT IT IS

Political capital encompasses the real capacity to influence decisions, regulate behavior and supervise systems. For citizens, it is their ability to demand accountability and contest decisions that affect them.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

A regulator without a technical tool can only declare principles without being able to verify them. That gap between declaration and real supervision is exactly where systems operate without accountability. Reactive post-tragedy regulation destroys more political capital than preventive action.

WHO RECEIVES THE DECISION

The citizen affected by an algorithmic decision has, in most cases, no political mechanism to contest it. They do not know what criteria were used and cannot identify who to complain to. The beneficiary loses their capacity to demand accountability over a decision they cannot name.

AFFECTED SECTORS

T4 — CRITICAL

| Emergency management · Energy infrastructure

T3 — HIGH RISK

| Judicial system · Migration

Public procurement · Public safety · Credit

WHY IT MATTERS — BOTH PERSPECTIVES

An immigrant denied residency by a risk algorithm does not just lose their case — they lose the ability to contest it because they cannot access the criteria that produced it. The official cannot defend or question it either. The political capital of both was neutralized by the same opacity.

ETHICAL

WHAT IT IS

Ethical capital is the moral trust that society places in an institution or professional. For the beneficiary, it is the expectation that the system considers them an equal and that decisions are guided by their wellbeing.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

The professional using opaque AI cannot guarantee their decision was fair. If the system produces biases — racial, gender, socioeconomic — the professional reproduces them unknowingly. Their ethical capital is compromised by a failure they did not design.

WHO RECEIVES THE DECISION

The beneficiary who receives a biased decision does not just lose the outcome: they lose the certainty that the system considers them an equal. Being reduced to a risk category without the right to an explanation is a form of indignity that no subsequent correction can fully undo.

AFFECTED SECTORS

T4 — CRITICAL

| Hospital triage and ICU · Emergency management

T3 — HIGH RISK

| Clinical diagnosis · Judicial system

Social protection · Public safety · Migration

WHY IT MATTERS — BOTH PERSPECTIVES

A criminal risk scoring algorithm that systematically assigns higher scores to people of certain backgrounds does not make a technical error — it commits an injustice. The judge does not know it. The accused cannot contest it. The ethical capital of the entire judicial system is compromised by a failure that neither of them can see.

TECHNOLOGICAL

WHAT IT IS

Technological capital encompasses algorithms, metrics, data and standards as proprietary assets. It includes the capacity to evaluate and demonstrate that systems work. A well-built technical standard is itself technological capital.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

Without verifiable standards, systems grow in power without growing in accountability. Organizations accumulate processing capacity but not the ability to demonstrate that what they process is trustworthy. That asymmetry increases the risk of catastrophic failure.

WHO RECEIVES THE DECISION

The beneficiary interacting with more powerful but less auditable systems faces a paradox: the technology becomes more opaque as it becomes more sophisticated. They have no tools to demand improvements in the dimensions that matter to them — accuracy, fairness, explainability.

AFFECTED SECTORS

T4 — CRITICAL

| Energy infrastructure · Traffic control · Water management

T3 — HIGH RISK

| Clinical diagnosis · Credit and banking · Pensions · Agriculture

WHY IT MATTERS — BOTH PERSPECTIVES

A company deploying AI in healthcare without a technical standard cannot demonstrate to its clients that the system is improving — or that it is not getting worse. The patient has no metrics to demand it. The physician has no criteria to evaluate it. A standard not only protects the beneficiary: it gives the organization the instrument to demonstrate that its technology deserves trust.

INSTITUTIONAL

WHAT IT IS

Institutional capital encompasses norms, stable procedures and the capacity to convert good practices into verifiable public policy. It is what makes an institution function predictably beyond the individuals who compose it at any given moment.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

An institution without an AI evaluation framework cannot demonstrate regulatory quality to international bodies or its own citizenry. The institutional vacuum in AI governance is a signal that other actors interpret as absence of technical capacity.

WHO RECEIVES THE DECISION

The citizen interacting with public institutions governed by AI without verifiable standards has no assurance that their rights are protected by a stable framework. The institutional structure that should provide legal security operates with systems that no one has certified.

AFFECTED SECTORS

T4 — CRITICAL

| Emergency management · Energy infrastructure · Water management

T3 — HIGH RISK

| Judicial system · Public procurement · Migration · Pensions · Education

WHY IT MATTERS — BOTH PERSPECTIVES

A citizen evaluated by an algorithmic system in a public institution has the right to have that evaluation comply with the principles the institution declares. If no technical standard verifies that compliance, the right exists on paper but not in practice. Institutional capital erodes every time a system operates without an architecture of accountability.

NATURAL

WHAT IT IS

Natural capital encompasses ecosystems, hydrological cycles, biodiversity, soil and climate. It is the only capital that cannot be substituted by other forms once destroyed and whose recovery operates on generational timescales.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

An AI operator managing natural resources can produce environmental damage without detecting it until the ecosystem can no longer recover. Without sustainability criteria in the standard, optimizing the system can mean destroying the environment.

WHO RECEIVES THE DECISION

Communities that depend on algorithmically managed ecosystems — farmers, indigenous communities, coastal zones — are the most vulnerable beneficiaries. They have no voice in system design, no mechanisms to contest its decisions, and bear consequences that may last for generations.

AFFECTED SECTORS

T4 — CRITICAL

| Water management · Energy infrastructure · Emergency management

T3 — HIGH RISK

| Agriculture and food security · Waste management · Territorial planning

WHY IT MATTERS — BOTH PERSPECTIVES

A community whose watershed is managed by AI optimized for energy efficiency — without territorial equity criteria — can lose water access without any system indicator recording it as a failure. The operator sees efficiency. The community sees scarcity. No standard that does not include natural capital as a governance variable can call itself complete.

INTELLECTUAL

WHAT IT IS

Intellectual capital encompasses codified knowledge, innovation capacity, intellectual property and the collective professional knowledge accumulated in a sector. It is what enables solving new problems with independent judgment.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

When AI replaces judgment without transparency, the knowledge the professional does not exercise atrophies. The intellectual capital of an entire profession can degrade within a generation if AI adoption does not preserve the exercise of professional judgment.

WHO RECEIVES THE DECISION

The beneficiary receiving care from a professional whose judgment has been gradually replaced by opaque AI receives, in the most complex cases, the response of a system not designed for that specific situation. The cases most in need of human judgment are exactly the ones algorithms handle worst.

AFFECTED SECTORS

T3 — HIGH RISK

| Clinical diagnosis · Laboratory

Educational assessment · Research · Pharmacy

WHY IT MATTERS — BOTH PERSPECTIVES

A medical resident who learns with AI without criteria for when to trust and when not to does not develop the differential judgment they will need in cases the algorithm cannot solve. In ten years, that specialty will have professionals faster in simple cases and less competent in complex ones — exactly the cases that matter most to the patient.

MANUFACTURED

WHAT IT IS

Manufactured capital encompasses physical infrastructure: hospitals, power plants, water networks, roads, bridges, tunnels, ports, airports and telecommunications. It is the material foundation on which everything else functions.

WHAT IS AT STAKE

WHO OPERATES THE SYSTEM

An operator delegating infrastructure decisions to AI without a resilience standard does not know under what conditions the system may fail in cascade. An algorithmic failure in a power grid or traffic control system is not a software error — it is a physical event.

WHO RECEIVES THE DECISION

The citizen who depends on infrastructure — the patient in a hospital whose power supply fails, the driver on an algorithmically managed road, the family without water — is the last link in a chain of failures that no one supervised with verifiable criteria.

AFFECTED SECTORS

T4 — CRITICAL

| Energy infrastructure · Water management · Traffic control · Roads and bridges · Critical telecommunications

T3 — HIGH RISK

| Urban mobility · Hospitals and health centers · Ports and airports

WHY IT MATTERS — BOTH PERSPECTIVES

A family that loses electricity because a grid management system failed algorithmically cannot file a technical complaint. The grid operator also cannot explain why the system made the decision it made. Manufactured capital is the most visible when it fails and the hardest to justify politically when the failure was produced by a system no one could audit.

THE CONCLUSION

**These eleven capitals are not abstract categories.
They are what professionals, institutions and
citizens
have built over years — and can lose
when an AI system fails without verifiable
guarantees.**

The harm does not fall only on those who operate the system.
It falls on those who receive its decisions — the patient,
the citizen, the community — who in most cases
do not know that an algorithm decided.

A technical standard protects both.

That is what still does not exist in most sectors.